MILITARY SPECIFICATION
PROJECTILE, 60MM, WP, XM722 BODY ASSEMBLY

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements not covered by the drawings and provides quality assurance provisions for the fabrication of the Projectile, 60MM, WP, XM722.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-63194	-	Propellant, M10 for 60mm Mortar M204 Propelling Charge

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	- Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-1949	- Inspection, Magnetic Particle

2.1.2 Other government documents, drawings and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER
(ARDEC)

9395680	- Projectile, 60MM, WP, XM722 Body Assembly
7548319	- Box Fiberboard Ammunition for Interplant Shipment of Projectile, Empty

(Copies of specifications, standards, drawings and publications and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E8	- Tension Testing of Metallic Materials
E10	- Brinnel Hardness of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B46.1	- Surface Texture

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all associated drawings and with all requirements specified in this specification. Threads shall be free of burrs which might interfere with assembly of the round. The cavity shall be free of all foreign matter.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied shall be uniform throughout each lot of projectiles and shall be performed and controlled so as to produce the mechanical properties specified on the applicable drawing. No part shall be re-heat treated if it fails to meet the applicable drawing requirements without Government approval.

3.4 Material defects.

3.4.1 Body. The body, including the cavity shall be free from cracks, splits, bursts, cold shuts, pipes, tears, blowholes, imbedded foreign matter shrinkage, porosity, inclusions, folds, seams and other similar defects when inspected by magnetic particle inspection or "white" (incandescent, flourescent, etc.) light.

3.4.2 Body cavity. In addition to the requirements of 3.4.1 the cavity shall be free from scale, fins, burrs, sharp edges, laminations, pits and other similar defects.

3.5 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI B46.1 shall be used as a basis for surface roughness determination.

3.6 Visual standards. Visual standards for evaluation of metal defects (3.4.2) will be established for each contract after the first month's production or whenever requested by the procuring activity. (See 6.1.1). However, the establishment of visual standards shall not waive the requirement for compliance with 3.4.1 and 3.4.2.

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3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Hydrotest. Each body shall successfully withstand a hydrostatic test load in accordance with dwg. 9392366 prior to machining the obturating band groove when tested in accordance with paragraphs 4.4.3.3 and 4.4.5.5 without cracking, leakage, rupture, or permanent deformation.

3.9 Proving ground (maximum average pressure). Projectile body assemblies shall reveal no evidence of failure when tested at maximum average pressure in accordance with 4.4.5.6.

3.10 Workmanship. All parts and assemblies shall be fabricated in a thorough workmanlike manner. Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings. Parts shall be free of burrs which might be injurious to personnel handling the item. No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no corrosion, grit, scale or other foreign matter remains therein after final cleaning. Projectile assemblies shall be prepared for painting in accordance with TT-C-490. The cleaning method used shall not be injurious to any part nor shall parts be contaminated by the cleaning agent. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shll be in accordance with the applicable specification. After painting, the body shall be free from bare spots, blisters, grease, dirt and other foreign matter. There shall be no pool of paint in the bottom of the cavity. Coating shall comply with the requirements of the applicable drawing. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provide equal assurance the provisions of this specification shall apply.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items as required per Table I, quantities are as follows:

Part description	Drawings	Quantity
Body, prior to obturating band groove	9392366	10
Body, before paint	9392366	10
Projectile, 60MM WP, XM722 - Assembly	9395680	30

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body, prior to machining obturating band groove (dwg. 9395680)				
	Hydrotest	10		3.8	4.4.3.3/4.4.5.5
	Body (dwg. 9392366)				
	Examination for defects	10		3.2	4.4.2.2
	Projectile, 60MM, XM722 Assembly (dwg. 9395680)				
	Salt spray test	5		3.2	4.4.5.4
	Examination for defects	5		3.2	4.4.2.3
	Maximum average pressure test	25		3.9	4.4.3.4/4.4.5.6
NOTES:					

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following:

4.4.1.1 Heat treatment batches. Heat treatment batches shall contain bodies from not more than one mill heat of steel, that have been heat treated in the same heat-treatment equipment, at the same temperature and time conditions in one unchanged process. No projectile shall be re-heat treated if it fails to meet the applicable drawing requirements without Government approval. The batch thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Projectile lots. Projectile lots shall consist of projectile assemblies produced by one manufacturer under one contract in one unchanged process, in accordance with the same drawing revision and specification revision, under conditions of a continuous production free of interruptions other than those due to the end of shift day or work week. All heat treatment batches that have complied with the mechanical properties requirements of the applicable component drawing may be re-grouped for assembly into projectile lots. A projectile lot may contain projectiles from more than one mill heat of steel.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

NOTE : Tool control dimensions. Dimensions marked tool control (see drawing) shall be gaged every two (2) hours or fraction thereof, during First Article Inspection, on the last component prior to a tooling change, adjustment or repair, and on the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body, Prior to Machining Obturating Band Groove	1 OF 1	9392366
			NEXT HIGHER ASSEMBLY: 9395680

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Hydrotest		100%	3.8	4.4.3.3/4.4.5.5
Major	None defined				
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.2	TITLE Body		SHEET 1 OF 3		DRAWING NUMBER 9392366
					NEXT HIGHER ASSEMBLY 9395680
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Radius at front of press fit surface improper or not continuous for 360° degrees		100%	3.2	Visual
2	Diameter of press fit surface incorrect		100%	3.2	Gage
3	Reverse taper (large diameter toward rear face) on press fit surface		100%	3.2	Gage
4	Press fit surface finish improper		100%	3.5	Visual/6.1
5	Magnetic particle test		1/	3.6	Visual/4.4.5.2
6	Hydrostatic test stamp missing		100%	3.2	Visual/4.4.3.3
Major					
101	Runout of press fit surface with bourrelet		0.40%	3.2	Gage
102	Runout of obturator groove with bourrelet		0.40%	3.2	Gage
103	Runout of ogive with bourrelet		0.40%	3.2	Gage
104	Runout of rear of body with bourrelet		0.40%	3.2	Gage
105	Runout of pitch diameter with bourelet and rear shoulder		0.40%	3.2	Gage
106	Diameter of obturator groove		0.40%	3.2	Gage
107	Width of obturator groove		0.40%	3.2	Gage
108	Thickness of wall at rear taper, (.125-. 030)		0.40%	3.2	Gage

NOTES:

1/ See 4.4.3.2

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.2	TITLE Body (cont'd)		SHEET 2 OF 3		DRAWING NUMBER 9392366
					NEXT HIGHER ASSEMBLY 9395680
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Thickness of wall at bourrelet		0.40%	3.2	Gage
110	Length from front face to rear shoulder		0.40%	3.2	Gage
111	Length from front face to rear of press fit surface		0.40%	3.2	Gage
112	Surface finish of cavity improper		0.40%	3.2	Visual
113	Diameter of bourrelet		0.40%	3.2	Gage
114	Runout of front face with bourrelet		0.40%	3.2	Gage
115	Weight		100%	3.2	Gage
116	Hardness	1/		3.2	Gage/4.4.5.1.1
117	Tension test	2/		3.2	Gage/4.4.5.1.2
118	Wall thickness at obturator groove		0.40%	3.2	Gage
119	Wall thickness at rear of obturator groove, min		0.40%	3.2	Gage
120	Surface rear of obturator groove not flush or below bourrelet		0.40%	3.2	Gage
121	Length, total, max		0.40%	3.2	Gage
122	Length from obturator groove shoulder to rear shoulder		0.40%	3.2	Gage
123	Profile of outside diameter incorrect		0.40%	3.2	Gage
124	Outside diameter at rear of thread		0.40%	3.2	Gage
125	Minimum effective thread length		0.40%	3.2	Gage
126	Distance from rear face to base of projectile cavity		0.40%	3.2	Gage
127	Length of thread boss		0.40%	3.2	Gage
128	Pitch diameter of thread		0.40%	3.2	Gage

NOTES: 1/ See 4.4.3.1.1
2/ See 4.4.3.1.2

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.2	Body (cont'd)		9392366
			NEXT HIGHER ASSEMBLY 9395680

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
129	Major diameter of thread, min		100%	3.2	Gage
130	Angle at rear of obturator groove		0.40%	3.2	Gage
131	Runout of pitch diameter with bourrelet and rear shoulder at projected distance		0.40%	3.2	Gage
Minor					
201	Length of thread undercut		0.65%	3.2	Gage
202	Diameter of thread undercut		0.65%	3.2	Gage
203	Depth of drill point, max.		0.65%	3.2	Gage
204	Radii of chamfers missing or incorrect		0.65%	3.2	Visual
205	Surface finish improper (except press-fit surface)		0.65%	3.5	Visual
206	Diameter of drill point, max		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile, 60MM, WP, XM722-Body Assembly	1 OF 1	9395680
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Surface finish of press fit surface incorrect (due to defective interior coating or paint		100%	3.2	Visual
2	Material defects (white light inspection)		100%	3.4	Visual/4.4.5.2.2
3	Proving ground test	1/		3.9	4.4.3.4/4.4.5.6
Major					
101	Diameter of bourrelet after painting		100%	3.2	Gage
102	Presence of paint or phosphate on threads or surfaces A, B, or C		0.40%	3.2	Visual
103	Corrosion preventative compound missing from threads, press fit surface or surfaces A, B, or C		0.40%	3.2	Visual
104	Paint fails salt spray test	2/		3.2	Visual/4.5.4
105	Presence of foreign matter in cavity		0.40%	3.2	Visual
106	Corrosion inhibitor missing from cavity prior to assembling protector cap		0.40%	3.2	Visual
107	Protective cap missing, damaged or improperly assembled		0.40%	3.2	Visual
Minor					
201	Location of marking improper		0.65%	3.2	Visual
202	Marking incorrect, incomplete, or illegible		0.65%	3.10	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1/ See 4.4.5.6
2/ See 4.4.5.4

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Box, Fiberboard, prior to sealing		7548319
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Support missing or damaged		0.65%	3.2	Visual
202	Separator missing or damaged		0.65%	3.2	Visual
203	Improper assembly		0.65%	3.2	Visual
204	Incorrect number on box		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.5	Box, Fiber, After Sealing	1	1	7548319
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR *100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Box damaged, cut or gouged through all layers		0.65%	3.2	Visual
202	Box not completely sealed		0.65%	3.2	Visual
203	Contents loose		0.65%	3.2	Visual/Manual
204	Marking missing, incorrect or illegible		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness test. Projectile shall be hardness tested to confirm uniformity of the heat treatment/stress relief operations and for the purpose of selecting the projectiles used for the tensile test. The first twenty-five (25) projectile bodies minimum, from the first heat treatment batch from one mill heat of steel (see 4.4.1.1) shall be subjected to this test. The hardest and softest projectiles from this group tested in accordance with 4.4.5.1.1 shall be sectioned for tensile testing. Projectiles from subsequent heat treatment batches (from the same mill heat of steel) shall use the same hardness range if the same time, temperature and process parameters are followed. Heat treatment batches whose parameters have deviated from established parameters shall require tensile tests of the hardest and softest from 25 deviating bodies as above. After the 25 are tested, testing shall be at the rate of 5 per hour. If the hardest or softest projectile from the group of 25 fails to meet the minimum tensile strength or elongation requirement, or if during the 5 per hour screening a harder than the hardest or softer than the softest projectile fails as above, then 100% hardness testing of the applicable heat treatment batch is required.

4.4.3.1.2 Tension test From the hardest and softest of the 25 projectiles submitted in accordance with 4.4.3.1.1, two tensile specimens shall be manufactured from each body, four total. When it has been determined that the applicable requirements for tensile strength and elongation have been met, the acceptable range for hardness shall be considered established for projectile batches from one mill heat of steel produced in accordance with 4.4.1.1. Should projectiles be found subsequently with a higher or lower hardness than in the established range, then a new set of tension test specimens must be prepared and tested in order to extend the acceptable hardness range. If the heat treatment or stress relief conditions of a batch have varied from designated process parameters, a separate set of tensile specimens for that batch shall be fabricated and hardness tests conducted as above. After projectiles from the segregated batch are qualified they may be combined with normal batches in a projectile lot (see 4.4.1.2).

4.4.3.2 Magnetic particle inspection. Testing shall be as specified in 4.4.5.2.1. After stress relief, hydrotesting and finish machining, bodies shall be inspected by magnetic particle inspection techniques approved by the responsible technical agency according to the following sequence:

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a. 100 percent magnetic particle inspection will be in effect until 2500 consecutive bodies have been inspected and found to be free of metal defects.

b. Reduced inspection may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten consecutive working days.

c. If no defect is found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive working days.

d. If no defect is encountered during the five days, the sample may be reduced to 5 bodies per hour.

e. If a defect is encountered by magnetic particle inspection during reduced sampling (b through d) the magnetic particle inspection shall revert back to (a) above. In addition a 100 percent back screening to the last acceptable hour's production will be performed.

f. For requalification of reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a through d) above.

g. Should a controversy arise as to the validity of an indicator, the manufacturer shall use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible technical agency.

h. Continual application of a 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as an alternative procedure in lieu of corrective action to eliminate a continuing material or process induced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdraw acceptance inspection until the contractor can prove his process and materials are capable of producing suspect free projectiles.

4.4.3.3 Hydrotest. After the wall thickness is in conformance with final size but prior to machining the obturating band groove, each projectile shall be subjected to an internal hydrotest in accordance with 4.4.5.5 to determine compliance with the requirements of 3.8. Each projectile that successfully passes the hydrotest shall be stamped "H" 0.125 inch high by .01 inch deep approximately 0.35 inch below the location of the zone used for projectile nomenclature immediately after removal from the hydrotest equipment.

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4.4.3.4. Providing ground (maximum average pressure) test. Samples provided in accordance with Table I and 4.4.2.3 will be subjected to this test. Defect classification and individual accept/reject criteria shall be in accordance with Table II. The test shall be conducted in accordance with 4.4.5.6.

TABLE II Proving Ground Test

Category	Defect	Accept/Reject	
Critical	Premature burst, metal part breakup or deformation due to launch forces	0	1
Major	Horizontal range less than 80% of average range	0	1

4.4.4 Inspection equipment. The inspection equipment required to Perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection-Method" column in the tables starting with paragraph 4.4.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.5 Methods of inspection.

4.4.5.1 Mechanical properties.

4.4.5.1.1 Hardness test. (See 4.4.3.4). The test shall be performed using any automated hardness tester which will produce accurate repeatable results with readings for an acceptable projectile between 25-80% of the full scale capability of the test equipment. When the Brinell hardness test is used equipment and procedures shall be as prescribed in ASTM Method E-10, except that the minimum dwell time shall be 5 seconds, otherwise operating procedures shall be as specified by the applicable equipment manual, standard ASTM test methods and the following subject to approval by the Government representative:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient decarburization zone. (Note: Impressions shall be completely removed by machining of the obturating band groove or final machining.)

b. The hardness tester calibration shall be tested at start up, at least once each four hours of continuous operation or after a layoff of three or more hours.

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c. The body shall be fixtured so that no rocking, shifting or deflection of the body will occur during the test.

d. The hardness tester shall not produce indentations deeper than 25% of the indentor ball diameter or wall thickness of the projectile in that zone.

e. Hardness readings shall be taken at least ten indentor ball diameters apart. At least two readings shall be taken per projectile and averaged in the zone where the obturating band groove will later be machined.

4.4.5.1.2 Tension testing. From each sample selected for testing in accordance with 4.4.3.1.2, two tension test specimens (one bourrelet area, one boat tail area) conforming to the largest obtainable standard or subsize flat or round specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body, 180 degrees apart. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements, and with the approval of the procuring agency. The Government shall not be restricted from testing additional specimens to determine the mechanical properties meet the drawing requirements. Procedures and criteria for determining yield strength, elongation, reduction in area and tensile strength shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0).

4.4.5.2 Metal defects inspection.

4.4.5.2.1 Before paint (Magnetic particle inspection). (See 4.4.3.2) Projectiles shall be tested using procedures for the wet fluorescent method as specified in MIL-STD-1949. Each projectile shall be tested twice. Both transverse and longitudinal fields shall be induced in the projectile using any method to establish the fields. Between the two (2) tests and following the second test, each projectile shall be demagnetized. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 mililters (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 roils of solids per 100 mls suspension when water is used. The suspension shall be checked at the start of each day’s production and at intervals of not over four hours of continuous operation, utilizing the procedures and equipment prescribed by section 4 of MIL-STD-1949. When performing the test for solid content of the suspension, if it is determined that it does not comply, all bodies

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that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of magnetic particle equipment personnel shall be qualified and certified in accordance with MIL- STD-410, Level II. After draining the excess suspension the body shall be inspected under ultraviolet light for the presence of defects. The requirement of MIL-STD-1949 to check system effectiveness by means of a steel ring specimen shall be waived. Components which exhibit indications which are suspected of being shallow surface defects may have material removed to a maximum of 0.005 inches subject to the following restrictions:

a. Involved areas shall be blended with adjacent surfaces to insure that no sharp edges, steps or gouges remain.

b. The involved areas shall be independently checked for verification that all drawing dimensions (wall thickness, variations, etc.) are within acceptable limits.

c. The components shall be again subjected to magnetic particle inspection for verification of defect removal.

4.4.5.2.2 After paint ("white" light). Each projectile shall be visually inspected internally and externally for the defects described in 3.4 using any type of lighting which will facilitate good inspection of all surfaces. Standards shall be used when available (see 3.6).

4.4.5.3 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of drawing 9395680 and the specification listed thereon.

4.4.5.4 Salt spray testing. Five salt spray samples (projectiles) shall be taken from each continuous final paint operation up to 24 hours long. One projectile shall be taken during the first 30 minutes of painting and the others at any other time. The test shall be conducted using component parts painted at the same time in the same equipment as other deliverable parts. The salt solution, equipment, preparation of test parts, test procedures and drying shall be in accordance with MIL-STD-810 Method 509.2 (salt fog). After drying, any evidence of paint failure, non-adhesion or corrosion on a surface painted with protective finish shall cause rejection of all parts painted during that operation. The duration of the test shall be as specified on drawing 9395680.

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4.4.5.5 Hydrotest. Projectile bodies shall be tested using equipment capable of performing the test accurately and consistently. During the test the pressure will be applied gradually until the required minimum pressure is reached and held for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a lay-off of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the test, the bodies tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Projectile rupture, cracking, leakage or bulging during the test shall be cause for rejection of the body. The test fixture shall not restrict the projectile body from expanding longitudinally or radially under pressure. Protective barriers should be installed where applicable to shield operators and material from metal fragments which may come from defective projectiles that fail the test (see 4.4.3.3).

4.4.5.6 Maximum average pressure test. The metal parts assemblies provided by Table I and 4.4.2.3 for this phase shall be inert loaded to the service weight and assembled with dummy fuzes into complete rounds. Firing tests will be conducted in a mortar for which the round is standard, the tube of which will have a minimum of 25 percent life remaining based on current mortar tube condemnation criteria. The propelling charge will be adjusted to obtain 109 to 117 percent of the maximum average pressure (10,080 psi) of the propellant when fired at 70°F. See MIL-P-63194. The test shall be performed at a Government proving ground in accordance with applicable U.S. Army TECOM acceptance test procedures. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.9 on the initial production sample, on the first five production lot samples, and on three (3) consecutive lots when a new mill heat of steel is introduced into production. An attempt will be made to recover at least 60 percent of the projectiles fired in this phase. The recovered projectiles will be examined to determine compliance with 3.9.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with requirements of dwg. 7548319.

6. NOTES

6.1 Visual examination qualifications. When compliance with the applicable classification of defect is in doubt as a result of visual examination, this characteristic shall be measured or gaged to determine acceptability.

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6.1.1 Visual standards. Visual standards to be established at contractor's plant in accordance with 3.6.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army ATTN: AMSMC-QAT-I (D) Bldg 62, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Submission of rejected magnetic particle test projectiles. The contractor shall submit five projectiles which fail the magnetic particle testing as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-B (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.5 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research Development and Engineering Center may also include drawings prepared by, and identified by as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

6.6 Subject term (key word) listing.

Defects, metal
Hardness
Hydro, test
Relief, stress
Strength, test
Taper, reverse
Test, magnetic particle
Test, maximum average pressure

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A424)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70679	PROJECTILE, 60MM, WP, XM722 BODY ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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